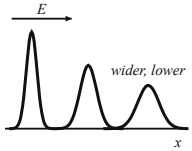
0 · Knob → what moves (p⁺-n: the light side is n)

- **Light-side** N ↑: V_0 ↑ (+60 mV/decade), W ↓ ($N^{-1/2}$), $\mathscr{E}_0$ ↑ ($N^{1/2}$), C_j ↑ ($N^{1/2}$), I_0 ↓ ($1/N$), V_{br} ↓ ($1/N$, toward Zener).
- **Heavy-side** N ↑: only V_0 moves (+60 mV/decade). W, C_j, I_0, V_{br} ≈ same.
- **Reverse** V_r ↑: W, $\mathscr{E}_0 \propto \sqrt{V_0 + V_r}$ ↑; C_j ↓; $I \approx -I_0$, flat.
- **Forward** V ↑: W, $\mathscr{E}_0$ ↓; C_j ↑; I, Δp_n, Q_p, C_s ×10 per 60 mV.
- T ↑: n_i^2 ↑↑ → I_0 ↑ (~15%/K); V at fixed I ↓ ~2 mV/K; V_0 ↓; V_{oc} ↓. Avalanche V_{br} ↑, Zener V_{br} ↓.
- E_g ↑: n_i ↓↓ → V_0 ↑, I_0 ↓↓, V_{oc} ↑, LED λ ↓.
- τ_p ↑: L_p ↑ (√), I_0 ↓ ($1/\sqrt{\tau}$), Q_p, C_s, t_{sd} ↑ ($\propto \tau$).
- **Area** ↑: I_0, I, C_j, C_s, $I_{sc} \propto A$. V_0, W, $\mathscr{E}$, V_{oc} unchanged.
- **Light ×10:** I_{sc} ×10, V_{oc} +60 mV. **Narrow base** ℓ ↓: $I \propto 1/\ell$ ↑, $t_t \propto \ell^2$ ↓.

1 · Haynes–Shockley, quasi-Fermi levels

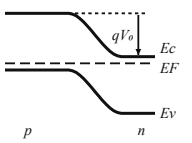
- Peak drifts: $\mu_p = \frac{d}{\mathscr{E} t_d}$. Spreads: $\Delta x = v_d \Delta t$, $D_p = \frac{(\Delta x)^2}{16 t_d}$. Area $\propto e^{-t/\tau_p}$.
- $\Delta x \propto \sqrt{t_d}$. Check $D/\mu = kT/q$.



- $n = n_i e^{(F_n - E_i)/kT}$, $p = n_i e^{(E_i - F_p)/kT}$, $np = n_i^2 e^{(F_n - F_p)/kT}$.
- Low level: the majority level barely moves; the minority level moves a lot.
- $J_n = \mu_n n \frac{dF_n}{dx}$: current needs a sloped quasi-Fermi level.

2 · Contact potential

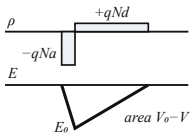
- $V_0 = \frac{kT}{q} \ln \frac{N_a N_d}{n_i^2}$ = (p side $E_i - E_F$) + (n side $E_F - E_i$).
- Si: $V_0 = 0.0596(\log N_a + \log N_d - 20.35)$.



- E_F flat; bands bend by qV_0; $\frac{p_p}{p_n} = \frac{n_n}{n_p} = e^{qV_0/kT}$.
- Drift = diffusion for each carrier separately; each is large.
- Typical Si: 0.6–0.9 V; always $< E_g/q$.

3 · Depletion region

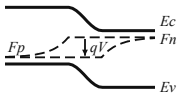
- $N_a x_{p0} = N_d x_{n0}$: the light side holds most of W. $x_{n0} = W \frac{N_a}{N_a + N_d}$.
- $W = \sqrt{\frac{2\epsilon(V_0 - V)}{q}\left(\frac{1}{N_a} + \frac{1}{N_d}\right)}$; one-sided $\sqrt{\frac{2\epsilon(V_0 - V)}{qN_{light}}}$.



- $\mathscr{E}_0 = \frac{qN_d x_{n0}}{\epsilon} = \frac{2(V_0 - V)}{W}$, peak at the junction, points n → p. Triangle area = $V_0 - V$.
- $\frac{V_n}{V_p} = \frac{x_{n0}}{x_{p0}} = \frac{N_a}{N_d}$: the light side drops most of it.
- Mobile carriers in W: essentially none.

4 · Bias, law of the junction

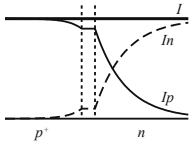
- Barrier → $V_0 - V$. Across W: $F_n - F_p = qV$.



- Edges: $\Delta p_n = p_n(e^{qV/kT} - 1)$, $\Delta n_p = n_p(e^{qV/kT} - 1)$. Reverse: → $-p_n$.
- Long side: $\delta p = \Delta p_n e^{-x_n/L_p}$, $L_p = \sqrt{D_p \tau_p}$.
- Low level needs $\Delta p_n \ll N_d$.

5 · Diode current and its components

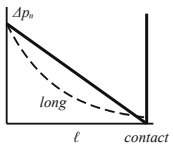
- $I = I_0(e^{qV/kT} - 1)$, $I_0 = qA\left(\frac{D_p}{L_p} p_n + \frac{D_n}{L_n} n_p\right)$.



- Forward: majority carriers cross the lowered barrier and become injected minority carriers. Reverse: minority carriers within L of W are swept across.
- The light side's minority carriers carry it (p⁺-n: holes).
- $I_p(x_n) = I_p(0)e^{-x_n/L_p}$; the total is constant; far away it is majority drift, $\mathscr{E} = \frac{J}{q\mu_n N_d}$ (tiny).
- ×10 per 60 mV. $e^{qV/nkT}$: $n \approx 2$ at low I (recombination in W) and at high I (high level); series R bends the top.
- Real reverse current grows with V_r: generation in W, $\propto W$.

6 · Narrow base ($\ell \ll L_p$)

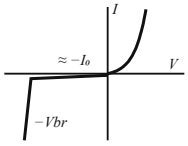
- Straight line, $\Delta p = 0$ at the contact. $I = qAD_p \frac{\Delta p_n}{\ell}$: $L_p \to \ell$, more current.



- $Q_p = \frac{1}{2} qA\ell \Delta p_n$, $t_t = \frac{Q_p}{I} = \frac{\ell^2}{2D_p} \ll \tau_p$: fast.
- Base drop $V_{base} \approx \frac{kT}{q} \frac{\Delta p_n}{N_d}$: an electron field cancels electron diffusion.

7 · Breakdown

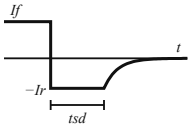
- $V_{br} = \frac{\epsilon \mathscr{E}_{crit}^2}{2qN} \propto 1/N$; $W_{br} = \frac{2V_{br}}{\mathscr{E}_{crit}}$.



- Zener (tunneling): heavy doping on both sides, $V_{br} < 4E_g/q \approx 4.5$ V, TC negative.
- Avalanche (impact ionization): $V_{br} > 6E_g/q \approx 6.7$ V, TC positive. Between: both, TC ≈ 0.
- $M = \frac{1}{1 - P}$; $P \to 1$ is breakdown. APD uses M as gain.

8 · Capacitance and switching

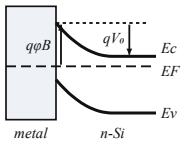
- $C_j = \frac{\epsilon A}{W} = A\sqrt{\frac{q\epsilon}{2(V_0 - V)} \frac{N_a N_d}{N_a + N_d}}$. Dominates in reverse.



- $\frac{1}{C_j^2} = \frac{2(V_0 + V_r)}{q\epsilon N A^2}$: slope → N_{light}, intercept at $V_r = -V_0$.
- $Q_p = I\tau_p$; $C_s = \frac{q}{kT} I\tau_p = G_s \tau_p$; $r = \frac{kT}{qI}$ (26 Ω at 1 mA). Dominates in forward.
- Turn-off: I flips to $-I_r$ and holds; V stays forward until $\Delta p_n = 0$. $t_{sd} \approx \tau_p \ln\left(1 + \frac{I_f}{I_r}\right)$.
- Faster: short τ_p (Au doping), narrow base, Schottky.

9 · Metal–semiconductor

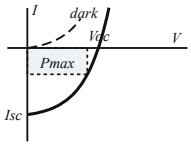
- n: $\phi_s = \chi + kT \ln \frac{N_c}{N_d}$. p: $\phi_s = \chi + E_g - kT \ln \frac{N_v}{N_a}$.
- n-type: $\phi_m > \phi_s$ rectifying, $\phi_m < \phi_s$ ohmic. p-type: the reverse.



- n: $\phi_B = \phi_m - \chi$ (from the metal, bias can't move it); $V_0 = \phi_m - \phi_s$ (from the Si, bias lowers it). p: $\phi_B = \chi + E_g - \phi_m$.
- Majority-carrier device: no stored charge, fast, lower turn-on. Practical ohmic: dope the surface heavily so carriers tunnel.

10 · Light: solar cell, photodiode, LED

- $I = I_{th}(e^{qV/kT} - 1) - I_{op}$, $I_{op} = qAg_{op}(L_p + L_n + W)$.



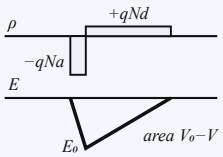
- $I_{sc} = I_{op}$; $V_{oc} = \frac{kT}{q} \ln\left(1 + \frac{I_{op}}{I_{th}}\right)$.
- $P_{max} = FF \cdot I_{sc} V_{oc}$, FF ≈ 0.7–0.85. Solar cell: 4th quadrant. Photodiode: 3rd (reverse), $I \propto$ light.
- PIN: wide i-region → collects more, low C, fast. APD: gain M.
- LED: $\lambda(\mu m) = 1.24/E_g(\text{eV})$, needs a direct gap. Laser: inversion ($F_n - F_p > h\nu$), a cavity, threshold current.

Mental math

- ×10 ↔ 59.6 mV; ×2 ↔ 18 mV; ×e ↔ 25.9 mV.
- $kT\ln(N/n_i)$: 10^{15} 0.288, 10^{16} 0.347, 10^{17} 0.407, 10^{18} 0.467 eV.
- $e^{V/kT}$: 0.5 V 2.42×10^{8}; 0.6 V 1.15×10^{10}; 0.7 V 5.47×10^{11}.
- $D=10$, $\tau=1\,\mu$s → $L=31.6\,\mu$m. $\sqrt{2}=1.41$, $\sqrt{10}=3.16$.

Ex 1 · Junction at equilibrium

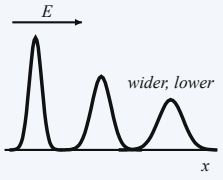
$N_a=4\times10^{17}$, $N_d=2\times10^{16}$.



- $V_0=0.0259\ln\frac{(4\times10^{17})(2\times10^{16})}{2.25\times10^{20}}=0.808$ V.
- $W=\sqrt{\frac{2(1.044\times10^{-12})(0.808)}{1.6\times10^{-19}}\left(\frac{1}{4\times10^{17}}+\frac{1}{2\times10^{16}}\right)}=0.235\,\mu$m; $x_{n0}=0.224\,\mu$m, $x_{p0}=11.2$ nm.
- $\mathscr{E}_0=2V_0/W=6.87\times10^{4}$ V/cm; $V_n=V_0\frac{N_a}{N_a+N_d}=0.770$ V.
- 5 V reverse: $W\times\sqrt{5.808/0.808}=0.631\,\mu$m; $\mathscr{E}_0=1.84\times10^{5}$ V/cm.

Ex 2 · Haynes–Shockley

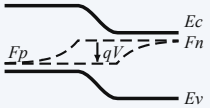
$\mathscr{E}=2$ V/cm, probe at 0.25 cm, $t_d=260\,\mu$s, $\Delta t=237\,\mu$s.



- $v_d=0.25/260\,\mu\text{s}=960$ cm/s → $\mu_p=480$ cm²/V·s.
- $\Delta x=960(237\times10^{-6})=0.228$ cm; $D_p=\frac{0.228^2}{16(260\times10^{-6})}=12.4$ cm²/s; $D/\mu=0.0259$ ✓.

Ex 3 · Edge carriers and quasi-Fermi levels

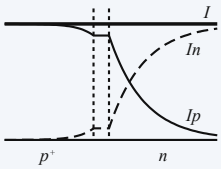
n side $N_d=5\times10^{15}$, $V=0.55$ V.



- $p_n=4.5\times10^{4}$; $e^{0.55/0.0259}=1.67\times10^{9}$; $\Delta p_n=7.51\times10^{13}$ (1.5% of N_d: low level).
- $F_n-E_i=0.0259\ln\frac{5\times10^{15}}{1.5\times10^{10}}=0.329$ eV; $E_i-F_p=0.55-0.329=0.221$ eV.

Ex 4 · Diode current and components

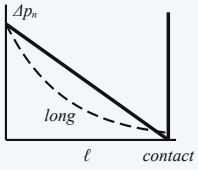
p⁺-n, $10^{18}/10^{16}$, $A=10^{-4}$ cm². n: $D_p=10$, $\tau_p=1\,\mu$s, $\mu_n=1250$. p: $D_n=6$, $\tau_n=0.1\,\mu$s. $V=0.6$ V.



- $L_p=31.6\,\mu$m, $L_n=7.75\,\mu$m. $I_0=qA\left(\frac{10(2.25\times10^{4})}{3.16\times10^{-3}}+\frac{6(225)}{7.75\times10^{-4}}\right)=1.17\times10^{-15}$ A.
- $I=I_0(1.15\times10^{10})=13.4\,\mu$A; holes $\frac{7.12}{7.29}=97.6\%$.
- Far field $\frac{J}{q\mu_n N_d}=\frac{0.134}{2.0}=0.067$ V/cm. For 10 mA: $0.0259\ln\frac{10^{-2}}{I_0}=0.771$ V.

Ex 5 · Narrow base

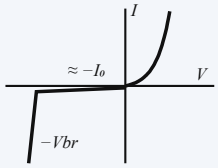
$\ell=5\,\mu$m, $N_d=10^{16}$, $D_p=10$, $A=10^{-4}$ cm², 0.55 V.



- $\Delta p_n=2.25\times10^{4}(1.67\times10^{9})=3.76\times10^{13}$.
- $I=qAD_p\frac{\Delta p_n}{\ell}=12.0\,\mu$A; $Q_p=\frac{1}{2}qA\ell\Delta p_n=1.50\times10^{-13}$ C.
- $t_t=\frac{\ell^2}{2D_p}=12.5$ ns; $V_{base}=(25.9\text{ mV})(0.00376)=97\,\mu$V.

Ex 6 · Breakdown

- p⁺-n, $N_d=10^{17}$, $\mathscr{E}_{crit}=5\times10^{5}$: $V_{br}=\frac{(1.044\times10^{-12})(2.5\times10^{11})}{2(1.6\times10^{-19})(10^{17})}=8.2$ V. Avalanche, just above 6.7 V.



- $N_d=10^{15}$, $\mathscr{E}_{crit}=3\times10^{5}$: $V_{br}=294$ V; $W_{br}=\frac{2V_{br}}{\mathscr{E}_{crit}}=19.6\,\mu$m.

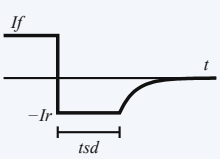
Ex 7 · Junction capacitance

p⁺-n, $10^{18}/10^{16}$, $A=10^{-3}$ cm².

- $V_0=0.814$ V. $C_j(0)=A\sqrt{\frac{q\epsilon}{2V_0}\frac{N_aN_d}{N_a+N_d}}=31.9$ pF.
- −5 V: $\times\sqrt{0.814/5.814}=11.9$ pF. +0.4 V: $\times\sqrt{0.814/0.414}=44.7$ pF.

Ex 8 · Stored charge and turn-off

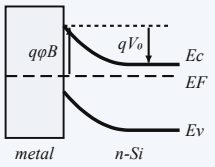
$\tau_p=1.5\,\mu$s, $I_f=2$ mA, switched to $I_r=1$ mA.



- $Q_p=I\tau_p=3.0$ nC; $C_s=\frac{Q_p}{kT/q}=116$ nF; $r=\frac{kT}{qI}=12.9\,\Omega$.
- $t_{sd}=1.5\ln(1+2)=1.65\,\mu$s.

Ex 9 · Two metals on n-Si

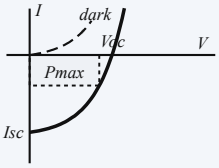
$N_d=10^{17}$, $\chi=4.05$ eV.



- $E_c-E_F=0.0259\ln280=0.146$ eV; $\phi_s=4.196$ eV.
- $\phi_m=4.6>\phi_s$: rectifying, $\phi_B=0.55$ eV, $V_0=0.40$ V.
- $\phi_m=4.0<\phi_s$: ohmic (electrons pile up at the surface).

Ex 10 · Solar cell

$A=2$ cm², $g_{op}=2\times10^{19}$, $L_p+L_n+W=46\,\mu$m, $I_{th}=10^{-11}$ A.



- $I_{sc}=qAg_{op}(46\times10^{-4})=29.4$ mA; $V_{oc}=0.0259\ln(2.94\times10^{9})=0.565$ V.
- $V_{mp}\approx0.487$ V: $P_{max}=13.6$ mW, FF = 0.82. ×10 light: $V_{oc}=0.624$ V.

Ex 11 · Light, fast

- GaN LED: $1.24/3.4=0.365\,\mu$m (UV). InP: $1.24/1.35=0.919\,\mu$m (IR).
- APD with $P=0.75$: $M=\frac{1}{0.25}=4$.